```
clc; clear; close all;

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
```

```
--- High-Pass Coefficients (Hanning Window) ---
```

```
disp(b_hann');
```

```
   -0.0000
         0
    0.0003
    0.0008
    0.0009
         0
   -0.0020
   -0.0038
   -0.0035
         0
    0.0057
    0.0100
    0.0087
         0
   -0.0127
   -0.0216
   -0.0183
         0
    0.0267
    0.0464
    0.0410
         0
   -0.0726
   -0.1568
   -0.2243
    0.7500
   -0.2243
   -0.1568
   -0.0726
         0
    0.0410
    0.0464
    0.0267
         0
   -0.0183
   -0.0216
   -0.0127
         0
    0.0087
    0.0100
    0.0057
         0
   -0.0035
   -0.0038
```

```
   -0.0020
         0
    0.0009
    0.0008
    0.0003
         0
   -0.0000
```

```
%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```